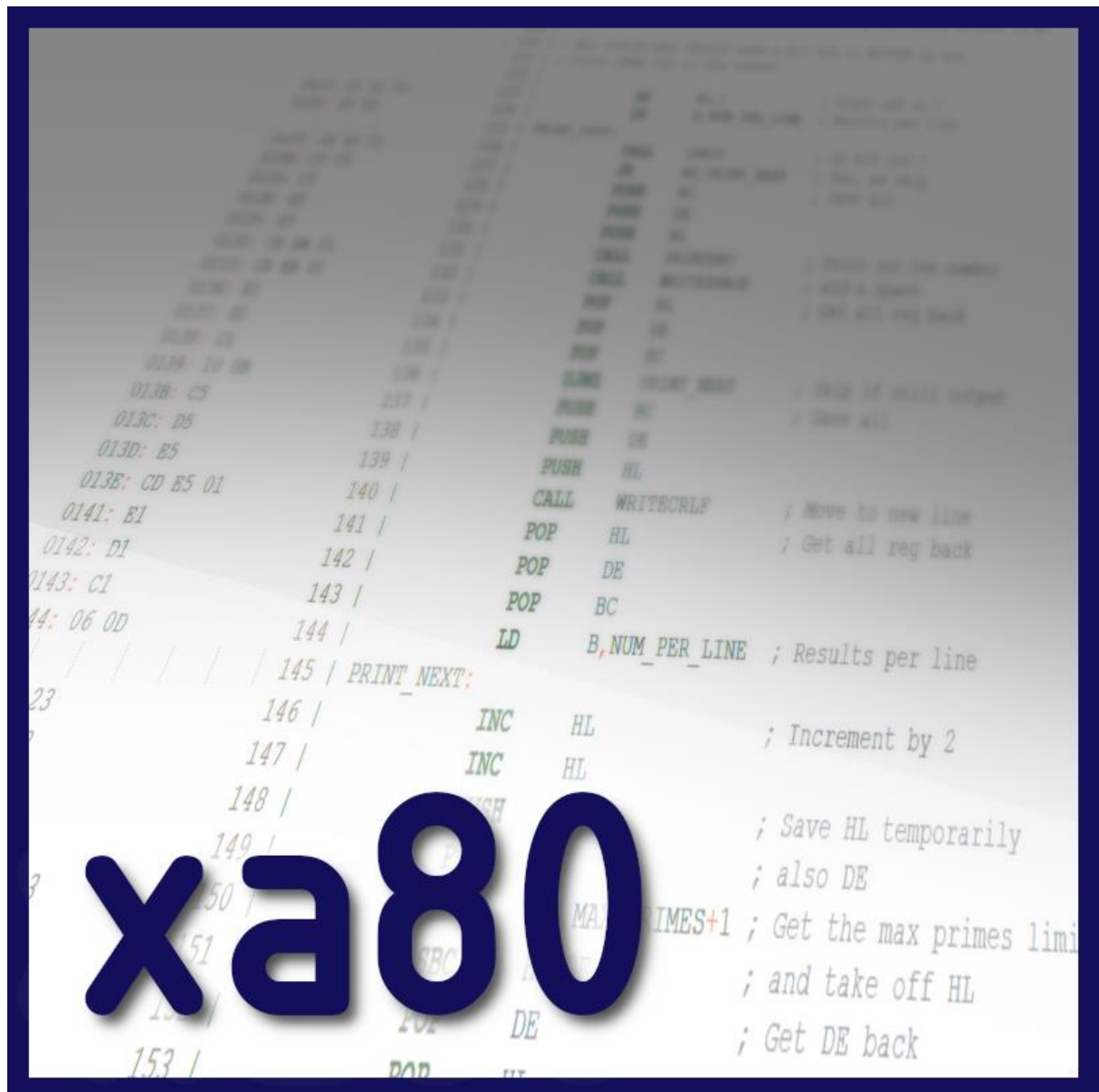


# xa80 Cross Assembler User Manual

## V1.0 (DEV)



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|         |             |                                     |
|         |             |                                     |
|         |             |                                     |

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<https://github.com/duncanamps/xa80>



## 1. Introduction

### 1.1. Disclaimer

xa80 is experimental open source software and is not guaranteed to work correctly in all conditions.

All trademarks are acknowledged as belonging to their respective owners.

### 1.2. Document purpose

This document is the User Manual for xa80, **x** (Cross) **a**sembler for **x80** processors. Its purpose is to provide a reference on how the application should be used, with examples where appropriate.

### 1.3. Application scope

xa80 is a multi-platform cross assembler available for Windows, Linux and MacOS. It is intended to be used with the following 8/16 bit processors:

- 8080
- 8085
- Z80<sup>1</sup>
- Z180

Being open source, the software naturally lends itself to being extensible should other processor types or families be required or other target operating systems be required.

### 1.4. Application features

Here are some of the key features of xa80:

- Open source
- Two pass assembler
- Supports mnemonics from different processors (8080, 8085, Z80, Z180)
- Opcode compiler so you can add your own secret/hidden instructions and extend to other processor variants in the "family"
- Macro capability with nested expansion of macros allowed
- Conditional assembly with IF / IFDEF / IFNDEF
- Full expression evaluator with many functions and string handling capability
- Rich set of command line switches
- xa80 Environment variable for commonly used parameters

---

<sup>1</sup> The Z80 is catered for as a standard to-the-book device, or the Z80X which allows the assembler to make use of the undocumented instructions

- Runs on any hardware supported by Lazarus/FPC (Windows, macOS, Linux, etc. etc.)
- Fast - will assemble the CP/M BDOS22.ASM (3,289 lines) and CCP22.ASM files (1,325 lines) with map file and listing outputs (total 105 pages) in approx 0.2 seconds using a Core i7 laptop, Acer Aspire 5 A515-56

## **1.5. Related documents**

Opcode compiler user manual

## 2. XA80 Quick Start

Here are some quick start examples to get you up to speed with XA80.

They assume the following:

- The examples have been downloaded from Github, not just the binaries
- The host system is Windows
- The target system is a CP/M based device with code starting at 0100 hex

### 2.1. Hello World for CP/M

At the command line, go to the folder **examples\hello\_z80** and run the following command:

```
xa80 hello.z80 --com --errorlog --hex --listing --map
```

A message similar to the following should appear:

```
XA80 Cross Assembler for x80 processors V1.0.0
Copyright (C)2020-2024 Duncan Munro

[ 0.015] Assembly started
[ 0.015] Processor is Z80
[ 0.031] Filename for com is
C:\Users\MyName\MyFolderChain\xa80\examples\hello_z80\hello.com
[ 0.031] Filename for listing is
C:\Users\MyName\MyFolderChain\xa80\examples\hello_z80\hello.lst
[ 0.031] Filename for hex is
C:\Users\MyName\MyFolderChain\xa80\examples\hello_z80\hello.hex
[ 0.031] Filename for map is
C:\Users\MyName\MyFolderChain\xa80\examples\hello_z80\hello.map
[ 0.031] Filename for error log is
C:\Users\MyName\MyFolderChain\xa80\examples\hello_z80\hello.log
[ 0.031] Assembling file
C:\Users\MyName\MyFolderChain\xa80\examples\hello_z80\hello.z80
[ 0.031] Assembly completed
```

A directory listing should display the following files:

```
Volume in drive C is Laptop
Volume Serial Number is 1AB2-47AA

Directory of C:\Users\MyName\MyFolderChain\xa80\examples\hello_z80

12/05/2024  12:59      <DIR>          .
12/05/2024  12:59      <DIR>          ..
12/05/2024  13:03             180 hello.com
12/05/2024  13:03             529 hello.hex
12/05/2024  13:03             117 hello.log
12/05/2024  13:03          1,950 hello.lst
12/05/2024  13:03             582 hello.map
25/01/2024  13:04             735 hello.z80
               6 File(s)              4,146 bytes
               2 Dir(s)  144,649,240,576 bytes free
```

Note that several new files have been created:

|                  |                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>hello.com</b> | The binary executable file for Hello World which CP/M can run                                                         |
| <b>hello.hex</b> | An text hex file containing the code from hello.com                                                                   |
| <b>hello.log</b> | A text file containing information on the success or otherwise of the assembly command                                |
| <b>hello.lst</b> | A detailed listing in text format of the source code and generated code                                               |
| <b>hello.map</b> | A text format map file which contains details of all the symbols and segments which were declared during the assembly |

Feel free to browse through the text files using your favourite editor to get an idea of the contents. You will learn much about the operation of xa80 by looking through them.

## 2.2. Multiple object files

@@@@@

### 3. xa80 General Structure

#### 3.1. File structure

The general structure of an assembly program is to have one or more source files which can be linked together as part of a larger piece of work.

Alternatively, a single source file could generate a **.COM** file or **.HEX** image which can be executed or transferred respectively.

#### 3.2. Include files

Optionally, include files can be referenced in source files using the **INCLUDE** directive and this can allow common definitions to be shared among a number of assembly source files.

#### 3.3. Segmentation

Starting with xa80 version 1.0, segments can be defined. This allows common groups of code or data to be combined later on as part of a linking process.

#### 3.4. Object files

Also starting with xa80 version 1.0, object files can be created. These describe the segments and labels used within an assembly, along with any external references. Debug information can also be included.

The purpose of object files is that they can be combined into a library and/or combined with other object files to create a larger piece of work.

## 4. xa80 Line Structure

The assembly is divided into a number of text lines, each of which is processed independently. The main elements of each line are:

- Labels
- Directives
- Instructions
- Macro references
- Operands
- Comments

### 4.1. Labels

The purpose of labels is to either identify a program location, or to define a variable value.

Labels are predominately alphabetic but can contain a number of allowed special characters. After the first character of the label, digits are also allowed.

The special characters are: . ? @ \_

Finally, the label may be suffixed by a colon : character, although this is optional. A popular convention is for program locations to use a colon and variable definitions to exclude this, however this convention is not enforced.

#### 4.1.1. Program location labels

Program location labels are used to access embedded data or to facilitate branch/jump/call instructions. They must always start on the first character of the line. A typical example would be:

```
START:                ; Program starts here
                      LD    B,4                ; Loop 4 times
                      LD    HL, TABLE         ; Point to data
LOOP:                 AND    A, (HL)           ; Mask bits
                      JR     Z, EXIT           ; Leave if done
                      INC    HL                ; Bump pointer
                      DJNZ   LOOP              ; Back round again
EXIT:
:
:
:
TABLE:               DB    0x1F, 0x23, 0x7A, 0xB7
```

### 4.1.2. Variable definition labels

Labels to define variables can be formed with either of the equate directives; these are **EQU** and **=**.

**EQU** would normally be used for a variable which is defined once and is unlikely to change during the assembly. If the value is re-assigned, assembly will continue, however a warning will be issued.

**=** has the same functionality as **EQU**, however no warning is issued when the value changes. This makes it well matched to looping structures such as **WHILE** and **REPEAT** which are covered in sections 6.20 and 6.16 respectively.

Labels defined as program locations can never be redefined and an error will be issued if this is attempted.

Some examples of variable definition labels:

|          |     |                   |                           |
|----------|-----|-------------------|---------------------------|
| MEMSIZE  | EQU | 8192              | ; Set program memory size |
| FAILMASK | EQU | %10110110         | ; Mask for fail bits      |
| MAX_RECS | EQU | 128               | ; Max records allowed     |
| REC_SIZE | EQU | 14                | ; Size of each record     |
| MEM_USE  | EQU | MAX_RECS*REC_SIZE | ; Calculated definition   |
|          |     | I = 0             | ; Reset loop var          |
|          |     | I = I + 1         | ; Increment loop var      |

#### Label case sensitivity

Labels are, by default, not case sensitive. This can be set by the environment variable and further overridden by the command line option **-k** or **--case-sensitive**. See sections 8.2.2 for more information on the environment and command line variables.

### 4.1.3. Using commands as labels

Some legacy code may use labels such as **WORD** which is an xa80 reserved word, a synonym for **DW**. In this instance, xa80 will allow the label to be used. A non-terminal warning will be issued.

However, the **WORD** command cannot be used subsequently in the assembly and this will generate a fatal error message if attempted.

Similarly, using any directive will prevent it being used as a label further on in the assembly.

Here are some examples:

```
WORD      EQU      2                ; Will generate a warning
LABEL     WORD     3                ; Error, WORD has been used as label

CRLF      DEFB     10,13           ; Use the DEFB directive
DEFB      LD       A,' ('         ; Error, DEFB has been used as cmd
```

## 4.2. Directives

Directives are the commands that drive the assembler behaviour which do not form part of the code itself. Directives may be followed by zero or more operands; this will depend on the nature of the directive itself.

More information on each of the directives and their use can be found in section 6.

Directives can be used “as is” or with a period preceding the directive, for example **EQU** and **.EQU** are interchangeable.

Some examples of directives are:

```
.ORG 0x0200                ; Set program addr
WARNOFF                    ; Turn warnings off
TITLE "Disk controller V2" ; Title of reports
.INCLUDE "macrodef.inc"    ; Include file
DB "Hello",13,10,"$"      ; Define some text
```

As directives can conditionally alter the assembly, it's not good practice to put a label on some directive lines as there is no guarantee the label will be handled correctly. Please review the following example:

```
DELAY      MACRO
            JR      NC, EXIT{#}
            LD      B, 18
D_LOOP{#}: NOP
            DJNZ    D_LOOP{#}
EXIT{#}:    ENDM
```

The macro, when expanded, will not include the first line **DELAY MACRO** or the last line **EXIT{#}: ENDM** lines so the **EXIT{#}** label will never be created.

But for some directives, a label is desirable or even essential, for example:

|        |    |                           |
|--------|----|---------------------------|
| TABLE: | DW | 0x12FA, 0x1410, 0x2AB8... |
|--------|----|---------------------------|

For this reason, the following behaviour is used:

| Directive example      | Label              | Examples                |
|------------------------|--------------------|-------------------------|
| Macro expansion        | Will cause Error   | ENDM                    |
| Not data defining      | Will cause Warning | ORG, IF, WHILE, WARNOFF |
| Data or macro defining | Accepted           | DB, DW                  |
| Macro definition       | Mandatory          | MACRO                   |

### 4.3. Instructions

Instructions are the backbone of the code generation. The general format is:

[label[:]] opcode [operand1 [, operand2 [, operand3]]] [comment]

The label will always be at the start of the line and may, optionally, be suffixed with a colon. The colon is not stored in the label list or referenced later on.

The opcode is one of the mnemonics defined for the chose processor type. Please note that the mnemonics for the Z80/Z180 are very different to the 8080/8085 although the functionality may be the same.

Whilst the opcodes for the referenced processors are set in stone, it's possible to use the opcode compiler to create extra instructions that are not present in the legacy processors. A separate document details the operation of the opcode compiler<sup>2</sup>.

There will normally be zero, one or two operands. When accessing undocumented instructions, a third operand may be used. For example, with **SET 0, [IX+disp], B** which means set bit 0 in the memory pointed to by IX plus a displacement, and then load that value into the B register.

A full list of opcodes available is detailed in appendix section 8.3.

<sup>2</sup> Amending the opcodes for existing processors will require that the assembler is recompiled to create a new binary. Adding new processors will also require amendments to the assembler source code

## 4.4. Macro references

Once a macro has been defined, it can be referenced simply by using the name of the macro. Optional operands may follow the macro name. Some examples of macro references are:

```

DELAY 18                ; Delay for 18 loops
MVMEM BUF, TABLE, 32  ; Move table to buffer
WRITE_STR              ; Write out a string to console

```

## 4.5. Operands

Operands are optionally used by directives, instructions and macro references. They can be registers, numeric or string, and a rich set of operators and functions allows calculations to be performed.

Expressions involving calculations are covered in more detail in section 5.

### 4.5.1. Register operands

Register operands are short and fixed definitions which typically refer to processor registers or flag conditions, the list is:

|      |           |      |
|------|-----------|------|
| A    | H         | NC   |
| AF   | HL        | NZ   |
| AF'  | (HL)      | P    |
| B    | I         | PE   |
| BC   | IX        | PO   |
| (BC) | (IX)      | PSW  |
| C    | (IX+disp) | R    |
| (C)  | IY        | SP   |
| D    | (IY)      | (SP) |
| DE   | (IY+disp) | Z    |
| (DE) | L         |      |
| E    | M         |      |

Not all operands are available on all processor types, for example PSW is available on 8080/8085, (IX) is available on Z80/Z180.

Any items from the table above, if used on a specific processor, effectively become reserved words. For example, **PSW** cannot be used as a variable in a 8085 assembler file as it's a register but it *could* be used in a Z80 assembler file.

Examples of register operands are:

```
LD    A,B
LD    HL,BC
LD    A, (HL)
LD    (IX+2),A
```

Some of the processor's use **(IX+n)** or **(IY+n)** where the intended address is a signed number added to the contents of the index register.

It is possible for the assembly file can contain these references *without* the displacement, for example:

```
OR    A, (IX+0)           ; The correct case, zero displacement
OR    A, (IX)             ; No displacement used
```

The first is the documented case, although the second example can alternatively be used. The second example will be assembled as if it were **OR A, (IX+0)** as there is no underlying Z80/Z180 code for **OR A, (IX)**.

## 4.5.2.Numeric operands

Numeric operands are 8 or 16 bit values which can be numeric constants, addresses, variables, or calculated values.

Numeric constants can be in binary, octal, decimal or hexadecimal forms. Examples are:

```
LD    A,%10110100        ; Binary
LD    B,0b10110100       ; Binary
LD    C,10110100B        ; Binary
LD    D,123O              ; Letter O, (not zero) for octal
LD    E,123Q              ; Letter Q also used for octal
LD    H,95                ; Decimal
LD    BC, #AB20           ; Hex
LD    DE,$AB20            ; Hex
LD    HL,0xAB20           ; Hex
LD    IX,0AB20H           ; Hex (must start with numeric!)
LD    IY,2AB8H            ; Hex
```

The use of letters in the constants is not case sensitive, **q** and **x** are treated the same as **Q** and **X**.

Addresses can be:

- Numeric operands
- Addresses which have been defined as program locations already
- Addresses which have yet to be defined

The following are examples of the above:

```
DEST:      DS      128                ; Buffer is 128 bytes
START:     LD      BC,0x0400          ; Number of bytes to copy
          LD      HL,TABLE            ; Get table address, defined later
          LD      DE,DEST             ; Get destination, already defined
TABLE:     DB      1,2,3,5,8,13,...
```

### 4.5.3.String operands

String operands can be enclosed within single or double quotes. If a numeric value is expected, the ordinal (ASCII) value of the first character of the string is taken. If the string is empty, then 0 is used.

Some examples are:

```
TITLE:     DB      "Hello World"
MESSAGE:   DB      'Error on line '
          LD      A,'0'                ; Offset to convert digit
```

Escape characters are permitted; the following are defined:

| Sequence | Result  | Notes                     |
|----------|---------|---------------------------|
| \a       | 07h BEL | Bell character            |
| \t       | 09h TAB | Tab character             |
| \n       | 0Ah LF  | Line Feed character       |
| \r       | 0Dh CR  | Carriage Return character |
| \e       | 1Bh ESC | Escape character          |
| \"       | 22h "   | Double quote              |
| \'       | 27h '   | Single quote              |
| \\       | 5Ch \   | Backslash                 |

### 4.5.4.Indirection

Some operands use indirection, for example:

```
LD      A,0x12                ; Load A with the value 0x12
LD      A,(0x12)              ; Load A with value stored in
                              ; memory location 0x0012
```

With parenthesis ( ), the instruction is taken to mean load from the address pointed to by the number. Without parenthesis, as in the first line, the instruction simply loads the literal value into the register.

The assembler also uses parenthesis to alter the precedence of results, however a set of rules allows precedence to be differentiated from indirection. The following examples demonstrate how this works in practice:

```
LD    A, (1+2*3+4)      ; Indirect
LD    A, (1+2) * (3+4)  ; Not indirect
LD    A, (IX+4)         ; Indirect
LD    A, (1+2) * 3      ; Not indirect
```

Please note that this feature is switched off for Intel (8080 / 8085) processors as the opcodes for these do not require or support indirection in the operand field.

## 4.6. Comments

Comments allow descriptive text to be added without influencing the operation of the assembler. There are three different types of comments available:

| Style | Format                   | Description                                                                                                                                             |
|-------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1a    | optional_text ; comment  | Any text from a ; onwards will be treated as a comment. Text prior to the ; will be treated as valid information and will be processed by the assembler |
| 1b    | optional text // comment | Any text from the // onwards will be treated as a comment                                                                                               |
| 2     | * comment                | A * at the start of a line will process all following characters as a comment                                                                           |

The following code example shows how comments can be used:

```
*****
*
*  ASSEMBLY FILE TEST
*
*****

BIT_MASK    EQU    01101001B      ; Use this to get correct flags
FACTOR      EQU    (10 + 3) * 2    // Calculation used

// Code starts here

START:
    XOR    A,A                    ; Zero A
    :      :
```



## 5. Expressions

Expressions can be integer or string in nature and are formed from literal values, symbols, operators and functions. Examples are:

```
A > B
1 << bit_5
2 + 3 * 4
LOW(address)
15 * (1 + 2)
Pos("-",title)
IIF(i>5,1,0)
build()
Left(title,3)
IIF(p==0,"Zero","Non-zero")
```

### 5.1. Literal values

Literal values can be:

- Binary numbers, prefixed by %, 0b or suffixed with B. For example %01101001, 0b11011 or 110B
- Octal numbers, suffixed with letter O or Q. For example 123O or 777q
- Decimal numbers – for example 123, 123D or 0
- Hexadecimal numbers, which can be prefixed by #, \$, 0x or suffixed by H<sup>3</sup>. For example #33A, \$ff78 or 33AH<sup>4</sup>
- String values enclosed in single or double quotes, for example "MyString"
- ASCII values of characters in single or double quotes, for example 'A' returns the hex value 65

### 5.2. Symbols

Symbols are constant values or variables used within the assembly. They can be associated with:

- A null value
- An integer value
- A string value

A null value is produced when a symbol is declared but has no specific value associated with it. It is of most use with the **DEFINED ()** function.

---

<sup>3</sup> For hex literals, and B/H suffixes these are not case sensitive

<sup>4</sup> Hex literals using the H suffix must start with a digit. This is to avoid confusion with labels as FABH could be a hex literal or a label. In this instance, use 0FABH to make it clear to the assembler that this is a literal value

### 5.3. Operators

The following table of operators has been defined in the assembler. Please see section 5.4 for details of the precedence used when calculating.

| Group   | Operator                                                  | Purpose                                                                                                                                                                                        | Notes                                                                                                                |
|---------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Math    | *<br>/<br>+<br>-<br>%<br>mod<br>( )                       | Multiply<br>Divide<br>Add<br>Subtract<br>Modulo<br>Grouping                                                                                                                                    | Both % and mod are synonymous<br>Use to group lower precedence items, e.g. 15*(2+3)                                  |
| String  | +                                                         | Concatenate                                                                                                                                                                                    | Adds two strings together                                                                                            |
| Bitwise | ~<br><br><<<br>shl<br><br>>><br>shr<br><br>&<br> <br>^    | Unary NOT<br><br>Shift left<br><br>Shift right<br><br>Bitwise AND<br>Bitwise OR<br>Bitwise XOR                                                                                                 | Operates in a 16 bit space, so ~0x1fff yields 0xe000<br>Both << and shl are synonyms<br>Both >> and shr are synonyms |
| Logical | !<br>&&<br>  <br>==<br>!=<br><><br><br><<br><=<br>><br>>= | Logical NOT<br>Logical AND<br>Logical OR<br>Compare equal<br>Compare not equal<br><br>Compare less than<br>Compare less than or equal<br>Compare greater than<br>Compare greater than or equal | != and <> are synonyms                                                                                               |

Comparison operators such as <= can be used on numeric values as well as strings. The only proviso is that you don't try and compare strings with numeric values as this

will trigger an error. String comparisons are case sensitive and this is not affected by the case sensitivity option for labels.

To compare strings in a case insensitive way, use **UPPER(a) <= UPPER(b)** for example.

Logical operators take inputs of 0 = False, anything else = True. They will return 0 for False or 1 for True. For example:

```
DB    123 || 7           ; Yields 1
DB    (1 > 2) && 17       ; Yields 0
DB    !123               ; Yields 0
```

## 5.4. Operator and Expression Precedence

Expressions are evaluated using the following precedence, lowest precedence number is evaluated first:

| Precedence | Type                      | Element                                      |
|------------|---------------------------|----------------------------------------------|
| 1          | Top of food chain         | ( bracketed expression )<br>Functions        |
| 2          | Unary expression          | +<br>-<br>! (logical not)<br>~ (bitwise not) |
| 3          | Multiplicative expression | *<br>/<br>%<br>mod                           |
| 4          | Additive expression       | +<br>-                                       |
| 5          | Shift expression          | <<<br>shl<br>>><br>shr                       |
| 6          | Compare expression        | <<br>><br><=<br>>=                           |
| 7          | Equivalence expression    | ==<br>!=<br><>                               |
| 8          | Binary AND expression     | &                                            |
| 9          | Binary XOR expression     | ^                                            |
| 10         | Binary OR expression      |                                              |
| 11         | Logical AND expression    | &&                                           |
| 12         | Logical OR expression     |                                              |

## 5.5. Integer Functions

These are functions returning an integer value. They may be dealing with strings.

| Function                           | Description                                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| ASC(string)                        | Takes the ASCII ordinal value of the first character of the string. If the string is empty, a value of zero is returned |
| DEFINED(variable)                  | Returns 1 if a variable has been defined or 0 if not                                                                    |
| HIGH(expression)                   | Returns the high byte of an expression (bits 8 to 15)                                                                   |
| IIF(expression,true_exp,false_exp) | If the expression is non-zero, true_exp is returned otherwise false_exp is returned                                     |
| LENGTH(string)                     | Returns the length of a string in characters                                                                            |
| LOW(expression)                    | Returns the low byte of an expression (bits 0 to 7)                                                                     |
| ORG()<br>\$                        | Returns the current program counter. \$ is a synonym for ORG()                                                          |
| POS(substr,string)                 | Returns the position of a substring within a string. If the substring is not found, zero is returned                    |
| VALUE(string)                      | Converts a string to a numeric value                                                                                    |

## 5.6. String Functions

A number of string functions are available within xa80:

| Function                                   | Description                                                                                                                            |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| BUILD()                                    | Provides the build number of the software as a string                                                                                  |
| CHR(expression)                            | Converts a numeric expression into an ASCII character. For example CHR(65) gives "A"                                                   |
| DATE()                                     | Return the date as a string in the form YYYY-MM-DD                                                                                     |
| HEX(expression)<br>HEX(expression, digits) | Returns a string of hex digits which represents the number. If digits is present, it is used to specify the minimum size of the result |
| IIF(expression,true_exp,false_exp)         | If integer expression is non-zero, the string expression true_exp is returned otherwise the                                            |

|                         |                                                             |
|-------------------------|-------------------------------------------------------------|
|                         | string expression false_exp is returned                     |
| LEFT(string,count)      | Take the leftmost count characters from a string            |
| LOWER(string)           | Take the lower case value of string                         |
| MID(string,start,count) | Take the middle of a string from start for count characters |
| PROCESSOR()             | Returns the processor selected for the assembly, e.g. "Z80" |
| STRING(number)          | Convert a number to a string value                          |
| RIGHT(string,count)     | Take the rightmost count characters from a string           |
| TIME()                  | Return the time as a string in the form HH:MM:SS            |
| UPPER(string)           | Return the upper case value of a string                     |
| VERSION()               | Version string for the assembler                            |

## 6. Directives

This section discusses the directives in more detail.

### 6.1. CPU

Ignored directive. It has been included for compatibility with earlier source code and raises a warning when used. xa80 is unable to support this directive as the processor type needs to be known at the start to create parsers, tokenisers, etc.

### 6.2. DB / DEFB – Define Bytes

The **DB** directive allows bytes of data to be defined in memory. These can come from 8 bit signed or unsigned values, or from string values. **DEFB** and **TEXT** are synonyms for **DB**.

Examples are:

```
TABLE:      DB      1,1,2,3,5,8,13,21...
HELLO:      DB      "Hello World"
COUNTL:    DB      LOW(BUFSIZE << 3)
COUNTH:     DB      HIGH(BUFSIZE<<3)
```

### 6.3. DC / DEFC – Define Characters

The **DC** directive operates identically to the **DB** directive, however strings are always encoded with bit 7 set of the last character in the string. **DEFC** is a synonym for **DC**.

For example:

```
KEYWRD:     DB      "END", "FOR", "NEXT", ...
```

Would be encoded as:

```

E  N  D  F  O  R  N  E  X  T
45 4E C4 46 4F D2 4E 45 58 D4 ...
```

Instead of:

```
45 4E 44 46 4F 52 4E 45 58 54 ...
```

Expressions yielding integers can be added in and these are not affected.

## 6.4. DS / DEFS – Define Storage

The **DS** directive allows storage to be allocated. The linker will reserve space in memory but will not store in the .com file.

Examples are:

```
BUF:      DS      128                ; Reserve 128 bytes
```

## 6.5. DW / DEFW – Define Word

The **DW** directive allows bytes of data to be defined in memory. These can come from 16 bit signed or unsigned values. **DEFW** is a synonym for **DW**. Data is stored in little-endian form, the low 8 bits is stored first.

Examples are:

```
TABLE:    DW      1,1,2,3,5,8,13,21...
TABLESIZE: DW      $ - TABLE
COUNT:   DB      BUFSIZE << 3
```

## 6.6. EXTERN

Not implemented at this time.

## 6.7. END

Defines the end of assembly. Any operands after the **END** directive will raise a warning. This directive is not actually needed to signal the end of the input.

## 6.8. EQU / = / SET - Equate

**EQU**ate allows a label to equate to a value. The value can be either a 16 bit integer or a string value, xa80 allows both.

**EQU** and **=** are similar, the only difference being that if a label is redefined with **EQU**, a warning is displayed. If the label is redefined with **=** then no warning is issued.

**SET** is like **=** however it is only available on Intel processors (8080 / 8085) as **SET** is already used as an opcode for Zilog processors.

Some examples:

```

START:                                ; Program label no warning
LABEL      EQU      10                ; No warning
J          = 7                      ; No warning
USER       EQU      "Fred"           ; No warning
NAME       =        "Bill"           ; No warning

START:                                ; Fatal error - redefining prog label
LABEL      EQU      12                ; Warning - redefining EQU label
J          = J + 1                    ; No warning, = allows this
USER       EQU      "John"           ; Warning - redefining EQU label
NAME       =        "Mike"           ; No warning, = allows this

```

## 6.9. GLOBAL

Not implemented at this time.

## 6.10. IF / ELSE / ENDIF

Allows conditional assembly. The format is:

```

IF      <expression>
<program block>
:
ENDIF

```

or:

```

IF      <expression>
<program block>
:
ELSE
<program block>
:
ENDIF

```

<expression> is any expression producing a logical value (zero or non-zero). Should the expression evaluate to 0 and the block is not assembled, it will still be necessary to part-assemble each line so that **ELSE** and **ENDIF** directives can be captured.

An example is:

```

IF      $ < 0xF800
DB      "Big long description of the assembler "
DB      "can be put in here",13,10,"$"
ELSE
DB      "Short description",13,10,"$"
ENDIF

```

When the input is assembled, the resulting listing file will put | at the sidebar to show code which has been included, and : to show code that has been excluded. This is helpful to follow the logic of **IF** / **ELSE** / **ENDIF**.

An example of the listing output is:

```

                                25 |    IF GOODVAL
                                26 |        MSGINFO "Producing some code"
0000: 00                        27 |        NOP
0001: 00                        28 |        NOP
0002: 00                        29 |        NOP
                                30 |    IF 1 > 2
                                31 :        MSGERROR "We shouldn't be here"
                                32 :    ELSE
0003: 3E 7B                    33 |        LD        A,123
                                34 |    ENDIF
                                35 |    ELSE
                                36 :        MSGERROR "We shouldn't be here"
                                37 :        LD        A, LOW(NEWVAL + 5678)
                                38 :    ENDIF
                                39 |
```

## 6.11. IFDEF / IFNDEF

These operate much like the IF command where the operand is a label name. IFDEF succeeds if the label is defined and IFNDEF if the label is not. The following lines are equivalent:

```
IFDEF table_base
IF DEFINED(table_base)
```

## 6.12. LISTOFF / LISTON

These directives take no operands and decide whether a listing output is produced or not. The directives can be interspersed at any point in the code.

## 6.13. MACRO / ENDM

Macros allow repetitive coding tasks to be represented as templates which can be deployed multiple times.

### 6.13.1. General format

The general format is:

```
<label>      MACRO <opt-param-names>
               <macro-definition>
               :
               ENDM
```

### 6.13.2. Optional parameter names

If parameters are going to be passed to the macro, the names can be listed during the **MACRO** line. These should follow the rules for naming labels, for example not starting off with a digit.

The parameter names are later expanded, using the { } characters.

### 6.13.3. Macro definition

The macro definition is like any other code block, however substitutions can be used either with a unique expansion serial number using {#} or the parameter with {param-name}.

### 6.13.4. ENDM

The ENDM directive must be present as it is the only way of terminating the macro definition.

### 6.13.5. Macro expansion

Macro expansion takes place when the label used to define the macro is later used as a directive. The following code gives an example of a simple macro definition and it being used.

```
ZERO4      MACRO address
            XOR    A,A                ; Zero A
            LD     [{address}],A
            LD     [{address}+1],A
            LD     [{address}+2],A
            LD     [{address}+3],A
            ENDM

            ZERO4 fpval1
```

### 6.13.6. Macro program labels

Program labels defined within macros are always global. Consider the following macro:

```
DELAY      MACRO
            LD      B, 18
D_LOOP:    NOP
            DJNZ    D_LOOP
            ENDM
```

Upon expanding the macro, one program label will be produced which is `D_LOOP`. This is all OK until you reference the macro a second time, it will create another `D_LOOP` variable which will cause an assembler error.

This is resolved by using the macro expansion serial number `{#}`. Each macro expansion will generate a sequentially increasing serial number which is guaranteed to be unique to that expansion. This can be embedded in the label, or indeed anywhere in the macro definition, to be substituted when the macro is expanded.

A rewrite of the previous example could be:

```
DELAY      MACRO
            LD      B, 18
D_LOOP{#}: NOP
            DJNZ    D_LOOP{#}
            ENDM
```

Using the macro reference `DELAY` twice in succession, will cause the following code to be assembled:

```
D_LOOP0:   LD      B, 18
            NOP
            DJNZ    D_LOOP0
            :
D_LOOP1:   LD      B, 18
            NOP
            DJNZ    D_LOOP1
```

Note that the labels are now unique to each macro expansion, so no error message will be generated.

### 6.13.7. Macro parameter expansion

Any parameter names supplied with the macro definition can be expanded. The passed parameter can be an integer value, a string, or even a register name.

An example of the parameters is:

```

DELAY      MACRO cycles
            LD      B,{cycles}          ; cycles contains the parameter
DLY2{#}:   NOP                        ; Short delay
            DJNZ DLY2{#}                ; Loop back if more to do

            DELAY 18                    ; Invoke the macro

```

Another example, this time using registers:

```

SWAP8      MACRO reg1, reg2            ; Swap 8 bit reg (but not A !)
            PUSH    AF                  ; Save A for now
            LD      A,{reg1}
            LD      {reg1},{reg2}
            LD      {reg2},A
            POP     AF

            SWAP8 H,L
            SWAP8 D,B

```

Macro parameters will match the case sensitivity of the assembler labels.

If case sensitivity is off (default or `--case-sensitive=0`) then a macro with named parameters of **Reg1** and **Reg2** can be expanded with `{REG1}`, `{reg1}` etc.

If case sensitivity is on (`--case-sensitive=1`) then a macro with named parameter of **Reg1** can only be expanded with `{Reg1}`. Using `{REG1}` or `{reg1}` will fail to expand.

### 6.13.8. Grouped macro parameters

Macro parameters can be grouped using the `<` and `>` characters to allow parameters to contain commas. The following gives an example of the use of grouped macro parameters:

```

mymacro    MACRO bcode, count
            DW      {count}
            DB      {bcode}
            ENDM

main       mymacro      <1,2,3,5,8,13,21>, 7

```

The above would expand to:

```

main       mymacro      <1,2,3,5,8,13,21>, 7
            DW      7
            DB      1,2,3,5,8,13,21

```

## 6.14. MSGINFO / MSGWARNING / MSGERROR

These provide messages on the console, or error log as the assembly takes place. They are only produced in pass 1 of the assembly.

The format is:

|            |                                   |
|------------|-----------------------------------|
| MSGINFO    | "Will show on console as info"    |
| MSGWARNING | "Will show on console as warning" |
| MSGERROR   | "Will show on console as error"   |

Use of MSGERROR will terminate the assembly process.

## 6.15. ORG

Sets the code origin for the assembler. The assembler starts assembling from address 0 and will increment as code and data bytes are output, unless this instruction is encountered.

An example is:

|     |        |                                            |
|-----|--------|--------------------------------------------|
| ORG | 0x0200 | ; Code starts at 0<br>; Set origin to 0200 |
|-----|--------|--------------------------------------------|

## 6.16. REPEAT / ENDR

**REPEAT** allows a block of code to be repeated a pre-defined number of times. For example, the macro example shown in section 6.13.5 could employ from the **REPEAT** directive:

|       |                  |          |
|-------|------------------|----------|
| ZERO4 | MACRO address    |          |
|       | XOR A,A          | ; Zero A |
|       | I = 0            |          |
|       | REPEAT 4         |          |
|       | LD [address+I],A |          |
|       | I = I + 1        |          |
|       | ENDR             |          |
|       | ENDM             |          |

## 6.17. SEGMENT

**SEGMENT** declares either a code or data segment which can be used later as part of an object library to group with similar segments generated from other source files. The general format is:

```
SEGMENT mandatory-name [, opt_modifier [, opt_modifier [...]]]
```

If code is generated with no preceding segment definition whatsoever, a default segment named **CSEG** will be created automatically.

### 6.17.1. Segment modifiers

The optional modifiers describe the segment and influence how it will be used later on at the linking stage.

These are:

| Feature on           | Feature off | Notes                                                                                                                                                                                                                                                     |
|----------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FIXED</b>         | RELOCATABLE | Decides whether a piece of code is fixed (the lowest generated address of code will be taken as the start address), or relocatable in which case the generated code will start from 0000 and end up at a different address once linked                    |
| <b>READONLY</b>      | READWRITE   | Decides whether a piece of memory can be overwritten or not. <i>Not currently implemented</i>                                                                                                                                                             |
| <b>UNINITIALISED</b> | INITIALISED | Uninitialised means memory is reserved, however no bytes are written to that area. The linker will simply put 00 bytes in that area; useful for variables. Initialised means that the memory contains defined bytes which could be zero or something else |
| <b>UNINITIALIZED</b> | INITIALIZED | US spelling synonyms for UNINITIALISED and INITIALISED, they do the same thing                                                                                                                                                                            |

'Feature off' is the default so:

```
SEGMENT CSEG          ; is exactly the same as...
SEGMENT CSEG, RELOCATABLE, READWRITE, INITIALISED
```

### 6.17.2. Segment syntactic checks

The assembler will check for lines which are syntactically incorrect such as:

```
SEGMENT CSEG, RELOCATABLE, FIXED      ; Clash of modifiers
SEGMENT CSEG, FIXED, FIXED            ; doesn't make sense
SEGMENT CSEG, FUBAR                   ; non-existent modifier
```

### 6.17.3. Segment semantic checks

Some combinations have limited practicality, for example READONLY and UNINITIALISED so a warning or error will be issued. The full list of combinations is:

| Movable     | Writable  | Initialised   | Priority       | Action          |
|-------------|-----------|---------------|----------------|-----------------|
| RELOCATABLE | READWRITE | UNINITIALISED | 4              |                 |
| RELOCATABLE | READWRITE | INITIALISED   | 3              |                 |
| RELOCATABLE | READONLY  | UNINITIALISED |                | Error – invalid |
| RELOCATABLE | READONLY  | INITIALISED   | 2              |                 |
| FIXED       | READWRITE | UNINITIALISED |                | Error – invalid |
| FIXED       | READWRITE | INITIALISED   | 1 <sup>+</sup> |                 |
| FIXED       | READONLY  | UNINITIALISED |                | Error – invalid |
| FIXED       | READONLY  | INITIALISED   | 1 <sup>+</sup> |                 |

<sup>+</sup>There can only be one fixed segment so all of these would be priority 1.

It is unlikely that a fixed segment would be uninitialized or empty, hence the error message.

Priority 1 would be loaded into memory first with priority 4 last.

### 6.17.4. Segment reintroduction

It may be necessary to bring a segment back into play, in which case just the name should be used. Any modifiers will be ignored on second and subsequent definitions, and a warning message will be issued if they are present.

Example of CSEG being reintroduced:

```
        SEGMENT CSEG, FIXED, READWRITE

        ORG    100H

START:   LD     HL, (vec_addr)
        LD     A, (HL)
        JP     FINAL

        SEGMENT DSEG, RELOCATABLE

vec_addr: DW    0                ; Vector address goes here

        SEGMENT CSEG            ; Reintroduce CSEG NO MODIFIERS!

FINAL:   RET     ; Back to caller
```

## 6.18. TITLE

Sets the title of the listing files, an example is:

```
TITLE "Disk controller V2"
```

## 6.19. WARNOFF / WARNON

These directives take no operands and decide whether warnings are produced or not. The directives can be interspersed at any point in the code.

They are overridden by the command line and environment in that if the environment/CL turns warnings off, the code cannot turn them on again.

## 6.20. WHILE / ENDW

The **WHILE** directive is very similar to the **REPEAT** directive. Unlike **REPEAT** where the expression is evaluated once at the start, the **WHILE** directive is evaluated every time it goes round the loop.

For this reason, it is possible for an infinite loop to be created.

An example of the **WHILE** directive is the following piece of code which creates a table of prime numbers:

```
;
; TEST_WHILE.Z80
;
; Test the WHILE and ENDW statements
;
; Duncan Munro
; 05/06/2023
;

;
; Generate a table of prime numbers up to 100
;

MAXPRIME    EQU    100

PRIMETABLE:
    J = 1
    WHILE J <= MAXPRIME
        MSGINFO "J=" + STRING(J)
        PRIME = 1
        K = 3
        WHILE PRIME && (K <= J / 2)
            MSGINFO "  K=" + STRING(K)
            IF (J MOD K) == 0
                PRIME = 0
            ELSE
                K = K + 2
            ENDIF
        ENDW
        IF PRIME
            DW J
        ENDIF
        J = J + 2
    ENDW

    PRIMECOUNT = ($ - PRIMETABLE) / 2

END
```

## 7. Companion Tools

The companion tools extend the functionality of the assembler by introducing a means to link together multiple object files into a program, or to group the object files in a library for easier retrieval.

### 7.1. Linker (xlink80)

@@@@@

### 7.2. Librarian (xlib80)

@@@@@



## **8. Appendices**

### **8.1. Appendix – Command Line**

The command line options are shown as follows:

XA80 Cross Assembler for x80 processors V1.0.0  
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Usage: xa80 filename(s) switches

Filenames can be discrete or wildcards, e.g. ..\source\\*.z80

Switches:

|         |                      |                                             |
|---------|----------------------|---------------------------------------------|
| -b <d>  | --debug=<d>          | Set the debug info inclusion level to <d>   |
| -c <cn> | --com=<cn>           | Set the .com output name to <cn>            |
| -d <id> | --define=<id>        | Define one or more symbols                  |
| -e <en> | --errorlog=<en>      | Set error log to <en>                       |
| -h      | --help               | Display help message                        |
| -i <il> | --include=<il>       | Set the include folders to <il>             |
| -k <o>  | --case-sensitive=<o> | Turn case sensitivity on or off (default 0) |
| -l <ln> | --listing=<ln>       | Set listing output file to <ln>             |
| -m <mn> | --map=<mn>           | Set map output file to <mn>                 |
| -o <on> | --object=<on>        | Set object output file to <on>              |
| -p <pt> | --processor=<pt>     | Set processor type to <pt>                  |
| -s <tp> | --show=<tp>          | Show more information on topic <tp>         |
| -t <ts> | --tab=<ts>           | Set tab size to <ts>                        |
| -v <n>  | --verbose=<n>        | Set verbosity to <n>                        |
| -w <o>  | --warnings=<o>       | Turn warnings on or off (default 1)         |
| -x <hn> | --hex=<hn>           | Set hex output file to <hn>                 |

<cn>, <en>, <ln>, <mn>, <on>, <hn> can be filenames or folders. If a folder, the output file will become the assembler file name with the filetype changed to .com/.log/.lst/.map/.obj80/.hex as appropriate. If a filename then this will be used for the output, if no filetype specified one will be added.

<b> is the debug info inclusion level and defines what gets added to the object

files, and can be:

- 0 No debug info inclusion (default)
- 1 Include variable names in the object file
- 2 Include source file line numbers and references along with variable names

<id> is the define and can be a symbol name, or symbol=value. 16 bit numbers and strings can be used. The ; separates multiple entries. For example:  
--define=DEBUG;PSIZE=128;TITLE="Home page"

<il> is the include list. Like with <id> ; can be used to separate.

<tp> topics are case insensitive and can be:

|              |                                                    |
|--------------|----------------------------------------------------|
| Distribution | Show distribution terms for this software          |
| Environment  | Show the environment for the assembler             |
| Instructions | Show the instruction set for the chosen processor  |
| Processors   | Show the different processors available by default |
| Reserved     | Show a list of reserved words                      |
| Version      | Show the version information for the software      |
| Warranty     | Show the warranty information for the software     |

<ts> sets the Tab size for tab expansions, the default is 4.

<n> can be one of:

- 0: Silent, only show fatal and internal software errors
- 1: Show only warnings and errors
- 2: Normal level, the default
- 3: Verbose, show more information
- 4: War and Peace, show much more information
- 5: Debug, only relevant with debug versions of the software

<o> can be one of the following boolean values:

- 0: Switches feature off
- 1: Switches feature on

## 8.2. Appendix - Environment

### 8.2.1.Defaults and precedence

The program environment is controlled by a series of defaults, the xa80 environment variable and the command line options.

In terms of precedence, the command line overrides the environment variable, and the environment variable overrides the defaults.

For example if the default for case sensitive is No, it can be overridden in the environment variable with

```
SET xa80=--case-sensitive=1;option;option;...
```

This will enable the assembler to respect case sensitivity on each use. However, the command line can override this.

```
xa80 source_files\*.asm --case-sensitive=0
```

The above will turn case sensitivity back off again.

### 8.2.2.Environment list

The following table lists the variables for the assembler and whether they can be amended in the environment or command line:

| Short | Long             | Values                                       | Default |
|-------|------------------|----------------------------------------------|---------|
| -b    | --debug          | Specifies the debug level                    | 0       |
| -c    | --com            | Specifies .com file                          | None    |
| -d    | --define         | Defines one or more symbols                  | Empty   |
| -e    | --errorlog       | Specifies the error log file                 | None    |
| -h    | --help           | Displays help on the program use             | N/A     |
| -i    | --include        | Sets the include folders to use              | Empty   |
| -k    | --case-sensitive | 0 = Not case sensitive<br>1 = Case sensitive | 0       |
| -l    | --listing        | Specifies the listing file                   | None    |
| -m    | --map            | Specifies the map file                       | None    |
| -o    | --object         | Specifies .obj80 file                        | None    |
| -p    | --processor      | 8080 = Intel 8080<br>8085 = Intel 8085       | Z80     |

| Short | Long       | Values                                                                                                                                                                                                                  | Default |
|-------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|       |            | Z80 = Zilog Z80<br>Z180 = Zilog Z180                                                                                                                                                                                    |         |
| -s    | --show     | Show the specified topic                                                                                                                                                                                                | N/A     |
| -t    | --tab      | Specifies the tab indent to use, typically 4 or 8                                                                                                                                                                       | 4       |
| -v    | --verbose  | 0 = Silent, only fatal errors<br>1 = Show only warnings and errors<br>2 = Normal informational level<br>3 = Verbose, show more info<br>4 = WarAndPeace, show lots of info<br>5 = Debug info, only used while developing | 2       |
| -w    | --warnings | 0 = Warnings off<br>1 = Warnings on                                                                                                                                                                                     | 1       |
| -x    | --hex      | Specifies the file to use for Intel hex listing                                                                                                                                                                         |         |

### 8.2.3. Specifying files

Filenames can be filenames, folders or the wildcard. They default to empty, so for example running the assembler with the following command line will not produce any output of any kind:

```
xa80 myfile.asm --processor=8080
```

#### 8.2.3.1. Filenames

A set filename can be used on the command line, although it's of limited use for the environment variable.

```
xa80 myfile.asm --hex=output
xa80 test.z80 --hex=myfile.hx2
```

The first line above will create the output file **output.hex** in the same folder as **myfile.asm** is located. Note that the **.hex** extension has automatically been added. If an extension is specified, as in the second line, it will not be overridden.

### 8.2.3.2. Folders

If the option obviously looks like a folder, it will be used and the output name will be made from the source assembly name with an appropriate extension. For example:

```
xa80 myfile.asm -c c:\temp\comfiles
```

If the folder `c:\temp\comfiles` exists, this will assemble `myfile.asm` into `c:\temp\comfiles\myfile.com` otherwise it will assemble it into `c:\temp\comfiles.com`.

To ensure the the parameter is recognised as a folder, put a suitable trailing delimiter for the operating system. For Linux or MacOS the following could be used:

```
xa80 myfile.asm -c /tmp/outputfiles/
```

If the folder `/tmp/outfiles/` does not exist, it will be created. The result file will be `/tmp/outputfiles/myfile.com`.

### 8.2.3.3. Wildcard output

It may be desirable to create an output file with the same base name as the input, in which case the following will carry this out:

```
xa80 myfile.asm --com --map
```

The above command will create the output files `myfile.com` and `myfile.map`.

You may also specify this by `*`.

```
xa80 myfile.asm --com=* --map=*
```

The example above has exactly the same outcome as the previous example. Finally, there is an option to use a wildcard with a different file extension. This is useful if you want to assemble a whole directory of `.asm` files but want, for example, a `.bin` file extension for the output:

```
xa80 *.asm --com=*.bin
```

### 8.2.3.4. Wildcard environment variable

If it's necessary to produce output files by default, these can be put in the environment variable by using just the \* character on its own:

```
SET xa80=--com=*;--map=*
```

The above will always create a suitably named .com and .map file as output.

### 8.2.4. Debug information

Debug information can be output, for example:

```
xa80 myfile.asm --object --debug=2
```

The following three options are valid:

| Option | Description                                                                                                                                                                                                                                                                                                              |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | No debug information is output from the assembly process. This will result in the smallest object file                                                                                                                                                                                                                   |
| 1      | All local symbols are exported. Normally local symbols will be exported if they have been marked as <b>GLOBAL</b> , or if they are cross-segment within the same module. This debug option allows all local symbols to be exported                                                                                       |
| 2      | All local symbols are exported as with option 1, however source line references are also exported. This will enable a suitable debugging application to load and display the original source lines. Please note that absolute file references are used so the debug information may not be useful on a different machine |

### 8.2.5. Symbol defines

Symbols can be predefined either from the command line or environment variable. Definition options are:

- Null value – symbol is defined but doesn't contain anything, useful with the **DEFINED()** function as it is set as defined with the value zero
- Numeric value, 16 bit signed or unsigned
- String value

Different fields are separated with the semicolon ; character, for example:

```
xa80 myfile.asm --define=DEBUG;BUFSIZE=128;TITLE="New prog"
```

There cannot be spaces between fields, however there can be spaces within strings.

### 8.2.6. Include folders

Much like symbol defines, a list of include folders can be submitted with folders separated by ; characters. Folders containing spaces must be enclosed by double quotes. For example:

```
xa80 myfile.asm -include=C:\temp;"c:\users\Duncan Munro"
```

### 8.2.7. Processor types

The list of available processor types can be increased by amending the software source code and compiling new opcode lists. This is an activity which is beyond the scope of this document.

### 8.2.8. Show topic

There are a number of "show" topics available from the assembler, some give useful information to the user, others are of most used when debugging the software. The full list of topics which can be displayed are:

| Topic        | Description                                                                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Distribution | Display the distribution terms for this software                                                                                                                  |
| Environment  | Shows the environment for the assembler and whether each environment option is a default, set by the xa80 environment variable, or overridden by the command line |
| Instructions | Shows the list of available instructions / mnemonics for the chosen processor. Best used with the -p / --processor option                                         |
| Operators    | Display a list of the operators and functions provided by the software                                                                                            |
| Processors   | Display a list of the processors supported by the assembler                                                                                                       |
| Reserved     | Display a complete list of reserved words that cannot be used for variables (this will vary depending on processor so use -p if needed)                           |
| Version      | Displays the version and build numbers of the software                                                                                                            |
| Warranty     | Displays the warranty provided by the software                                                                                                                    |

The topics are not case sensitive. For example

```
xa80 --show=Version  
xa80 --show=RESERVED --processor=8080
```

### 8.3. Appendix – Opcodes

The following opcodes are defined by the application:

| Opcode | 8080 | 8085 | Z80 | Z180 |
|--------|------|------|-----|------|
| ACI    | Y    | Y    |     |      |
| ADC    | Y    | Y    | Y   | Y    |
| ADD    | Y    | Y    | Y   | Y    |
| ADI    | Y    | Y    |     |      |
| ANA    | Y    | Y    |     |      |
| AND    |      |      | Y   | Y    |
| ANI    | Y    | Y    |     |      |
| BIT    |      |      | Y   | Y    |
| CALL   | Y    | Y    | Y   | Y    |
| CC     | Y    | Y    |     |      |
| CCF    |      |      | Y   | Y    |
| CM     | Y    | Y    |     |      |
| CMA    | Y    | Y    |     |      |
| CMC    | Y    | Y    |     |      |
| CMP    | Y    | Y    |     |      |
| CNC    | Y    | Y    |     |      |
| CNZ    | Y    | Y    |     |      |
| CP     | Y    | Y    | Y   | Y    |
| CPD    |      |      | Y   | Y    |
| CPDR   |      |      | Y   | Y    |
| CPE    | Y    | Y    |     |      |
| CPI    | Y    | Y    | Y   | Y    |
| CPIR   |      |      | Y   | Y    |
| CPL    |      |      | Y   | Y    |
| CPO    | Y    | Y    |     |      |
| CZ     | Y    | Y    |     |      |
| DAA    | Y    | Y    | Y   | Y    |
| DAD    | Y    | Y    |     |      |
| DCR    | Y    | Y    |     |      |
| DCX    | Y    | Y    |     |      |
| DEC    |      |      | Y   | Y    |
| DI     | Y    | Y    | Y   | Y    |
| DJNZ   |      |      | Y   | Y    |
| EI     | Y    | Y    | Y   | Y    |
| EX     |      |      | Y   | Y    |
| EXX    |      |      | Y   | Y    |
| HALT   |      |      | Y   | Y    |
| HLT    | Y    | Y    |     |      |
| IM     |      |      | Y   | Y    |
| IN     | Y    | Y    | Y   | Y    |
| INO    |      |      |     | Y    |

| Opcode | 8080 | 8085 | Z80 | Z180 |
|--------|------|------|-----|------|
| INC    |      |      | Y   | Y    |
| IND    |      |      | Y   | Y    |
| INDR   |      |      | Y   | Y    |
| INI    |      |      | Y   | Y    |
| INIR   |      |      | Y   | Y    |
| INR    | Y    | Y    |     |      |
| INX    | Y    | Y    |     |      |
| JC     | Y    | Y    |     |      |
| JM     | Y    | Y    |     |      |
| JMP    | Y    | Y    |     |      |
| JNC    | Y    | Y    |     |      |
| JNZ    | Y    | Y    |     |      |
| JP     | Y    | Y    | Y   | Y    |
| JPE    | Y    | Y    |     |      |
| JPO    | Y    | Y    |     |      |
| JR     |      |      | Y   | Y    |
| JZ     | Y    | Y    |     |      |
| LD     |      |      | Y   | Y    |
| LDA    | Y    | Y    |     |      |
| LDAX   | Y    | Y    |     |      |
| LDD    |      |      | Y   | Y    |
| LDDR   |      |      | Y   | Y    |
| LDI    |      |      | Y   | Y    |
| LDIR   |      |      | Y   | Y    |
| LHLD   | Y    | Y    |     |      |
| LXI    | Y    | Y    |     |      |
| MOV    | Y    | Y    |     |      |
| MLT    |      |      |     | Y    |
| MVI    | Y    | Y    |     |      |
| NEG    |      |      | Y   | Y    |
| NOP    | Y    | Y    | Y   | Y    |
| OR     |      |      | Y   | Y    |
| ORA    | Y    | Y    |     |      |
| ORI    | Y    | Y    |     |      |
| OTD    |      |      |     | Y    |
| OTDM   |      |      |     | Y    |
| OTDMR  |      |      |     | Y    |
| OTDR   |      |      | Y   | Y    |
| OTI    |      |      |     | Y    |
| OTIM   |      |      |     | Y    |
| OTIMR  |      |      |     | Y    |

| Opcode | 8080 | 8085 | Z80 | Z180 |
|--------|------|------|-----|------|
| OTIR   |      |      | Y   | Y    |
| OUT    | Y    | Y    | Y   | Y    |
| OUT0   |      |      |     | Y    |
| OUTD   |      |      | Y   | Y    |
| OUTI   |      |      | Y   | Y    |
| PCHL   | Y    | Y    |     |      |
| POP    | Y    | Y    | Y   | Y    |
| PUSH   | Y    | Y    | Y   | Y    |
| RAL    | Y    | Y    |     |      |
| RAR    | Y    | Y    |     |      |
| RC     | Y    | Y    |     |      |
| RES    |      |      | Y   | Y    |
| RET    | Y    | Y    | Y   | Y    |
| RETI   |      |      | Y   | Y    |
| RETN   |      |      | Y   | Y    |
| RIM    |      | Y    |     |      |
| RL     |      |      | Y   | Y    |
| RLA    |      |      | Y   | Y    |
| RLC    | Y    | Y    | Y   | Y    |
| RLCA   |      |      | Y   | Y    |
| RLD    |      |      | Y   | Y    |
| RM     | Y    | Y    |     |      |
| RNC    | Y    | Y    |     |      |
| RNZ    | Y    | Y    |     |      |
| RP     | Y    | Y    |     |      |
| RPE    | Y    | Y    |     |      |
| RPO    | Y    | Y    |     |      |
| RR     |      |      | Y   | Y    |
| RRA    |      |      | Y   | Y    |
| RRC    | Y    | Y    | Y   | Y    |

| Opcode | 8080 | 8085 | Z80 | Z180 |
|--------|------|------|-----|------|
| RRCA   |      |      | Y   | Y    |
| RRD    |      |      | Y   | Y    |
| RST    | Y    | Y    | Y   | Y    |
| RZ     | Y    | Y    |     |      |
| SBB    | Y    | Y    |     |      |
| SBC    |      |      | Y   | Y    |
| SBI    | Y    | Y    |     |      |
| SCF    |      |      | Y   | Y    |
| SET    |      |      | Y   | Y    |
| SHLD   | Y    | Y    |     |      |
| SIM    |      | Y    |     |      |
| SLA    |      |      | Y   | Y    |
| SLP    |      |      |     | Y    |
| SPHL   | Y    | Y    |     |      |
| SRA    |      |      | Y   | Y    |
| SRL    |      |      | Y   | Y    |
| STA    | Y    | Y    |     |      |
| STAX   | Y    | Y    |     |      |
| STC    | Y    | Y    |     |      |
| SUB    | Y    | Y    | Y   | Y    |
| SUI    | Y    | Y    |     |      |
| TST    |      |      |     | Y    |
| XCHG   | Y    | Y    |     |      |
| XOR    |      |      | Y   | Y    |
| XRA    | Y    | Y    |     |      |
| XRI    | Y    | Y    |     |      |
| XTHL   | Y    | Y    |     |      |

## 8.4. Appendix – Error Codes

### 8.4.1. Informational codes

| Code  | Text                                            | Notes                                                                                                                                                                                                 |
|-------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I0000 | <user message>                                  | Generated in response to the MSGINFO directive described in section 6.14                                                                                                                              |
| I0001 | Assembly started                                |                                                                                                                                                                                                       |
| I0002 | Assembly completed                              |                                                                                                                                                                                                       |
| I0003 | Assembling file <filename>                      |                                                                                                                                                                                                       |
| I0004 | Filename for <output type> is <filename>        | Used to describe the hex output file name, com file name, etc.                                                                                                                                        |
| I0005 | Processor is <proctype>                         | Information and describes the processor type for which the assembly is being made                                                                                                                     |
| I0006 | Searching for include file <name> at <location> |                                                                                                                                                                                                       |
| I0007 | Processing include file <name>                  |                                                                                                                                                                                                       |
| I9999 | <debug message>                                 | Issued to the console or log when debug messages are enabled. Refer to section 8.2.2 for setting verbosity levels. Note that debug messages cannot be produced by production releases of the software |

### 8.4.2. Warning codes

| Code  | Text                                                  | Notes                                                                                                          |
|-------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| W1000 | <user warning>                                        | Generated in response to the MSGWARN directive described in section 6.14                                       |
| W1001 | Code wrapped round back to zero                       | Occurs if address FFFFh has been filled in a segment causing the origin counter to go back to zero             |
| W1002 | Directive <name> ignored                              |                                                                                                                |
| W1003 | Symbol <identifier> has been redefined                |                                                                                                                |
| W1004 | Operands after END directive ignored                  | Some assemblers allow operands after an END directive, e.g. END MYFILE however xa80 ignores them               |
| W1005 | Symbol <identifier> is undefined                      |                                                                                                                |
| W1006 | Symbol <identifier> replaces command of the same name | It's possible to repurpose commands as symbols in certain circumstances, however this will result in a warning |
| W1007 | Macro parameter count                                 | Caused by calling a macro with the                                                                             |

|              |                                                                       |                                                                                                                                                                                                                                                |
|--------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | mismatch - expected <number> parameters, received <number>            | wrong number of parameters                                                                                                                                                                                                                     |
| <b>W1008</b> | Outputting code with no segment definition, default CSEG created      | V1.0 onwards requires at least one segment to operate correctly. If a code generating line is encountered with no segment, a default will be created. Use the SEGMENT command to explicitly set up a segment and avoid this warning            |
| <b>W1009</b> | Segment modifiers ignored, segment <segname> has already been defined | On the first definition of a segment, you may put specifiers such as Fixed, ReadOnly, etc. On subsequent declarations of the same segment name, it will not be possible to use specifiers and they will be ignored                             |
| <b>W1010</b> | Segment modifier <modifier> clashes with a preceding modifier         | Appears when conflicting or opposing modifiers are present, e.g. SEGMENT Fixed,Relocatable                                                                                                                                                     |
| <b>W1011</b> | .COM file is empty                                                    | A .COM file has been specified on the command line, however no code has been generated and the resulting .COM file will have zero bytes                                                                                                        |
| <b>W1012</b> | .HEX file is empty                                                    | A .HEX file has been specified on the command line, however no code has been generated and the resulting .HEX file will not declare any bytes                                                                                                  |
| <b>W1013</b> | ORG command is forcing relocatable segment <segname> to become Fixed  | If a segment has been declared as relocatable, any use of an ORG command will force the segment to have a fixed attribute                                                                                                                      |
| <b>W1014</b> | Unresolvable value                                                    | The expression parser has encountered operations on an external reference or relocatable internal address. For example if an external address PRINT has been declared, a statement such as DW PRINT<<3 will come up with an unresolvable value |
| <b>W1015</b> | Segment modifiers are confusing                                       | An attempt to combine segment modifiers which make no sense. For example, combining READONLY with UNINITIALISED or FIXED with UNINITIALISED                                                                                                    |

### 8.4.3. Error codes

| Code  | Text                                                                    | Notes                                                                                                                    |
|-------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| E2000 | <user error>                                                            | Generated in response to the MSGERROR directive described in section 6.14                                                |
| E2001 | Illegal escape character <char>, valid are <valid_list>                 |                                                                                                                          |
| E2002 | Unterminated string <text>                                              |                                                                                                                          |
| E2003 | Unrecognised content <text>                                             |                                                                                                                          |
| E2004 | Expected number <text>                                                  |                                                                                                                          |
| E2005 | Integer overflow                                                        |                                                                                                                          |
| E2006 | Binary literal <text> is too short                                      |                                                                                                                          |
| E2007 | Octal literal <text> is too short                                       |                                                                                                                          |
| E2008 | Hex literal <text> is too short                                         |                                                                                                                          |
| E2009 | Divide by zero                                                          |                                                                                                                          |
| E2010 | Expected string <text>                                                  |                                                                                                                          |
| E2011 | Expected positive number <text>                                         |                                                                                                                          |
| E2012 | String <text> failed to convert                                         |                                                                                                                          |
| E2013 | Parser error <text>                                                     |                                                                                                                          |
| E2014 | Unable to parse input <text>                                            |                                                                                                                          |
| E2015 | Code symbol <text> has already been defined                             |                                                                                                                          |
| E2016 | Mandatory colon not present in code label <text>                        | Only of use if mandatory colons for labels are specified. This is likely to be removed in later versions of the software |
| E2017 | Unexpected operands                                                     | A directive had operands when operands were not expected                                                                 |
| E2018 | Operand no. <number> is of indeterminate data type                      |                                                                                                                          |
| E2019 | Expected integer                                                        |                                                                                                                          |
| E2020 | Unexpected label <text>, ignored                                        |                                                                                                                          |
| E2021 | Instruction not available for <text>                                    |                                                                                                                          |
| E2022 | Operands expected                                                       |                                                                                                                          |
| E2023 | Byte must be in range - 127..255                                        |                                                                                                                          |
| E2024 | Code buffer limit of <number> bytes exceeded                            |                                                                                                                          |
| E2025 | Empty string is not allowed                                             |                                                                                                                          |
| E2026 | Integer must be in the range <number> to <number>                       |                                                                                                                          |
| E2027 | Illegal distance of <number> for relative branch, should be - 128..+127 |                                                                                                                          |
| E2028 | Bit number for SET/RES must                                             |                                                                                                                          |

|              |                                                                   |                                                             |
|--------------|-------------------------------------------------------------------|-------------------------------------------------------------|
|              | be in the range 0..7                                              |                                                             |
| <b>E2029</b> | Operand for IM instruction must be in the range 0..2              |                                                             |
| <b>E2030</b> | Using reserved word for label <text>                              |                                                             |
| <b>E2031</b> | File not found <text>                                             |                                                             |
| <b>E2032</b> | Unexpected character <text> in input                              |                                                             |
| <b>E2033</b> | Undefined parser table action for state <number> and token <text> |                                                             |
| <b>E2034</b> | Unexpected token <text> in input                                  |                                                             |
| <b>E2035</b> | PARSER_STACK_SIZE_MAX (<number>) exceeded                         |                                                             |
| <b>E2036</b> | Invalid character in label                                        |                                                             |
| <b>E2037</b> | Attempt to perform activities after END directive                 |                                                             |
| <b>E2038</b> | Invalid option <text> for -s/--show command line parameter        |                                                             |
| <b>E2039</b> | Mandatory value missing after switch <text>                       |                                                             |
| <b>E2040</b> | Equals expected but not found in command or environment           |                                                             |
| <b>E2041</b> | Premature end of string in command or environment <text>          |                                                             |
| <b>E2042</b> | Invalid command line switch <text>                                |                                                             |
| <b>E2043</b> | <text> is not a valid command directive or processor instruction  |                                                             |
| <b>E2044</b> | Include file <text> not found                                     |                                                             |
| <b>E2045</b> | Maximum number of includes (<number>) exceeded                    |                                                             |
| <b>E2046</b> | Expected label, found <text>                                      |                                                             |
| <b>E2047</b> | Unexpected end: <text>                                            |                                                             |
| <b>E2048</b> | Unexpected ENDIF                                                  |                                                             |
| <b>E2049</b> | Unexpected ELSE                                                   |                                                             |
| <b>E2050</b> | More than one ELSE statement between IF and ENDIF                 |                                                             |
| <b>E2051</b> | Unexpected ENDW                                                   |                                                             |
| <b>E2052</b> | ENDW in different file to WHILE statement (<text>)                |                                                             |
| <b>E2053</b> | Unexpected ENDR                                                   |                                                             |
| <b>E2054</b> | ENDR in different file to REPEAT statement (<text>)               |                                                             |
| <b>E2055</b> | Unexpected ENDM                                                   | Typically occurs if ENDM is encountered when not in a macro |

|              |                                                                          |                                                                                                                  |
|--------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
|              |                                                                          | definition                                                                                                       |
| <b>E2056</b> | Cannot define a macro within another macro                               | Nesting of macros at the definition stage is not allowed, however a macro may contain another (predefined) macro |
| <b>E2057</b> | Macro <text> not found                                                   |                                                                                                                  |
| <b>E2058</b> | Cannot place a label on an ENDM directive                                |                                                                                                                  |
| <b>E2059</b> | Command line define error, could not process <text>                      |                                                                                                                  |
| <b>E2060</b> | Unexpected directive <text>, have already processed directive for <text> |                                                                                                                  |
| <b>E2061</b> | Could not load processor details from <text>                             |                                                                                                                  |
| <b>E2062</b> | Command <text> has already been used as a label                          |                                                                                                                  |
| <b>E2063</b> | Command <text> cannot now be used as a label                             |                                                                                                                  |
| <b>E2064</b> | Failed macro expansion                                                   |                                                                                                                  |
| <b>E2065</b> | Illegal segment modifier <text>                                          |                                                                                                                  |
| <b>E2066</b> | Cannot create .COM file as only 1 fixed segment must be present          |                                                                                                                  |
| <b>E2067</b> | Cannot create .HEX file as only 1 fixed segment must be present          |                                                                                                                  |
| <b>E2068</b> | Cannot change Relocatable segment to Fixed after code has been generated |                                                                                                                  |
| <b>E2069</b> | Global symbol <text> not found in this module                            |                                                                                                                  |
| <b>E2070</b> | External symbol <text> has already been defined locally                  |                                                                                                                  |
| <b>E2071</b> | Attempt to generate code in an uninitialised segment                     |                                                                                                                  |

#### 8.4.4. Internal Error codes

These occur when an error exists within the software; specifically, something has happened internally which should not have happened.

These are not typically caused by a user mistake.

Codes are numbered X3001 to X3018 and specific text will be displayed to allow the finding to be reported. There is also a catch all X3999 which is an unhandled exception error.

| Code  | Text                                                                | Notes |
|-------|---------------------------------------------------------------------|-------|
| x3001 | Unhandled case option at <routine>                                  |       |
| x3002 | Preparser peek error                                                |       |
| x3003 | Could not find procedure <procedure> in grammar                     |       |
| x3004 | Reduction code not defined for rule no. <index> (<rulename>)        |       |
| x3005 | Binary constant <string> failed to convert                          |       |
| x3006 | Hex constant <string> failed to convert                             |       |
| x3007 | Invalid element type                                                |       |
| x3008 | Resource not found <name>                                           |       |
| x3009 | Parser tables not loaded                                            |       |
| x3010 | Unexpected goto from table                                          |       |
| x3011 | Stack empty                                                         |       |
| x3012 | Goto expected for rule index #<index> but not found in table        |       |
| x3013 | Attempt to set lexer buffer size while in the middle of an activity |       |
| x3014 | Attempt to pop from empty execution stack                           |       |
| x3015 | Unhandled expression source                                         |       |
| x3016 | Unhandled scope                                                     |       |
| x3017 | Symbol "<symbol>" not propagated correctly                          |       |
| x3999 | Unhandled exception                                                 |       |

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